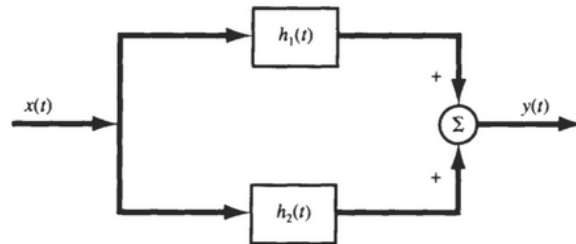


Q.1 The system shown in figure is formed by connection two systems in parallel. The impulse response of the systems are given by $h_1(t) = e^{-2t}u(t)$ and $h_2(t) = 2e^{-t}u(t)$. Find the impulse response $h(t)$ of the overall system.



Q.2 The impulse response of a discrete time LTI system is given by $h[n] = (\frac{1}{2})^n u[n]$. Let $y[n]$ be the output of the system with the input $x[n] = 2\delta[n] + \delta[n - 3]$. Find $y[1]$ and $y[4]$.